

Vansh

Kapurthala, Punjab, India | vanshbhutani2005@gmail.com | +91-7888845986
[LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Portfolio](#)

Professional Summary

Computer Science undergraduate (CGPA 8.65/10, IKGPTU, 2027) specializing in Machine Learning, Generative AI, and real-world deployed systems. Experienced in building production-ready ML pipelines, RAG systems, and explainable AI models. Seeking AI/ML or Data Science internships.

Technical Skills

Languages	Python, SQL, C++, JavaScript
Machine Learning / Deep Learning	Scikit-learn, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Classification, Regression, Neural Networks, Transformers, Finetuning
Generative AI	RAG, LangChain, LLM Integration, Prompt Engineering, Embeddings
MLOps & Deployment	FastAPI, Flask, Docker, Git, REST APIs, Model Deployment
Data	Pandas, NumPy, Matplotlib, EDA, Data Preprocessing, NLP, Time Series Analysis
Explainability	SHAP, Model Interpretability, Model Reliability Testing

Projects

PranRakshak AI — Intelligent Hospital Command Center for Early Sepsis Detection [Live Demo](#) | [GitHub](#)

- Built a real-time sepsis early-warning system using the PhysioNet CinC 2019 ICU dataset with Sepsis-3 clinical criteria.
- Implemented SHAP-based explainability to identify key risk-driving features for interpretable clinical predictions.
- Developed a Retrieval-Augmented Generation (RAG) clinical copilot supporting natural-language queries and OCR-based lab report ingestion.
- Deployed a production-ready full-stack system using FastAPI backend, PostgreSQL database, and React frontend.
- Actively improving model reliability, feature engineering, and deployment scalability.

Quora Question Pairs — NLP Semantic Similarity Detection [Demo](#) | [GitHub](#)

- Processed 400K+ question pairs and engineered TF-IDF and semantic similarity features.
- Achieved 79% accuracy and F1-score of 0.70 using classical ML models.
- Evaluated transformer-based sentence embeddings to benchmark performance improvements.

Customer Churn Prediction — Artificial Neural Network Model [Demo](#) | [GitHub](#)

- Designed and trained an ANN model on structured telecom customer data.
- Achieved 87.73% test accuracy and reduced validation loss through hyperparameter tuning.
- Compared performance with Logistic Regression and Random Forest models.

Student Performance Prediction — End-to-End Machine Learning Pipeline [Demo](#) | [GitHub](#)

- Built a modular ML pipeline including data ingestion, preprocessing, training, evaluation, and deployment.
- Achieved 88.05% prediction accuracy and deployed the model using Flask REST API.
- Implemented structured logging, exception handling, and automated model evaluation.

Experience

AI & Machine Learning Intern — EduNet Foundation (AICTE + IBM) Jun 2025 – Aug 2025

- Completed full ML lifecycle including preprocessing, training, and deployment.
- Evaluated models using accuracy, F1-score, and cross-validation.

Education

B.Tech in Computer Science and Engineering — IK Gujral Punjab Technical University 2023 – 2027

CGPA: 8.65 / 10

Achievements

- Finalist — Hack On Hills, NIT Hamirpur
- Top 5 Finalist — LPU Hackathon
- 3rd Prize — University Startup Competition

Leadership

- Founder — Upstarts (Technology & Startup Club)
- Class Representative — CS Department

Certifications

- IBM — Fundamentals of Artificial Intelligence
- Anthropic — AI Fluency Certification

Research Interests

Generative AI · Agentic AI · LLM Systems · Retrieval-Augmented Generation (RAG) · AI Agents · Model Deployment · Explainable AI · Machine Learning Systems